

Highlights

Brain-Aligned Multi-Stream Video Transformers with Sparse Self-Selection

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- Introduces a neuro-inspired split-and-fuse video transformer with separate "what" and "where" streams whose complementary EEG alignment patterns are consistent with ventral and dorsal visual pathway specializations.
- Replaces dense self-attention with a sparse winner-takes-all selection module, inspired by competitive routing observed in biological visual circuits, for efficient video processing.
- Achieves competitive or state-of-the-art accuracy and pareto frontier accuracy-latency trade-offs on Kinetics-400 and Something-Something V2 among models of comparable scale and training.
- Demonstrates improved robustness to spatial perturbations compared to strong video transformer baselines.
- Shows higher EEG representational similarity than existing video models, indicating more brain-aligned visual representations.

Brain-Aligned Multi-Stream Video Transformers with Sparse Self-Selection

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ABSTRACT

Modern video transformers typically ignore principles from primate vision and are rarely evaluated against neural data, limiting their biological interpretability. We introduce a sparse winner-takes-all token selection module that replaces dense self-attention to improve efficiency and approximate competitive routing observed in biological visual circuits. We further propose a neuro-inspired split-and-fuse video transformer which uses two complementary pathways: a high-resolution, low-frame-rate "what" stream and a low-resolution, high-frame-rate "where" stream, fused before classification. On Kinetics-400 and Something-Something V2, our best variant operates on the Pareto frontier of accuracy versus inference time among models of comparable scale and pretraining, and showing improved robustness to spatial perturbations. Using representational similarity analysis between model embeddings and time-resolved EEG recordings for the same video stimuli, our model attains a peak brain-model correlation of 0.18 (about 78% of the noise ceiling) and consistently outperforms strong video transformer baselines, suggesting that pathway specialization and sparse competition are useful inductive biases for efficient, brain-aligned video understanding.

1. Introduction

Modern video understanding models have achieved impressive performance on large-scale benchmarks, yet they are typically developed with little regard for how biological visual systems process dynamic scenes (Yamins and DiCarlo, 2016; Celeghin, Borriero, Orsenigo, Diano, Méndez Guerrero, Perotti, Petri and Tamietto, 2023). Existing transformer-based architectures for video recognition often rely on a single stream with dense self-attention over all space-time tokens, which offers strong expressivity but leads to quadratic scaling in sequence length and limited interpretability in relation to primate vision (Bertasius, Wang and Torresani, 2021; Arnab, Dehghani, Heigold, Sun, Lučić and Schmid, 2021; Liu, Ning, Cao, Wei, Zhang, Lin and Hu, 2022b). At the same time, a growing NeuroAI literature suggests that models whose internal representations align better with neural data can exhibit improved robustness and may provide more mechanistic insights into perception, but systematic evaluations of video transformers against human brain activity remain scarce (Schrimpf, Kubilius, Hong, Majaj, Rajalingham, Issa, Kar, Bashivan, Prescott-Roy, Geiger et al., 2018; Dapello, Kar, Schrimpf, Geary, Ferguson, Cox and DiCarlo, 2023; Conwell, Prince, Kay, Alvarez and Konkle, 2024).

In contrast, primate vision relies on multiple specialized pathways that process complementary aspects of the visual input. The ventral "what" pathway is associated with detailed object recognition, whereas the dorsal "where" pathway emphasizes motion, spatial relations, and action-related information (Goodale and Milner, 1992; Creem and Proffitt, 2001). This division of labor suggests that explicit pathway specialization could serve as a useful inductive bias for artificial systems that must understand complex video streams (Simonyan and Zisserman, 2014; Zareh, Toulabinejad, Manshaei and Zahabi, 2024). However, most current video transformers process inputs in a single pathway and do not exploit a structured separation between high-resolution spatial processing and high-temporal-resolution motion processing (Bertasius et al., 2021; Liu et al., 2022b).

Another gap concerns computational efficiency and biological plausibility in attention mechanisms. Standard self-attention forms a dense weighted sum over all tokens for each query, which is computationally expensive and difficult to reconcile with evidence for sparse, competitive interactions in neural circuits (Itti and Koch, 2001). Neurophysiology and computational models point to winner-takes-all dynamics and lateral inhibition, in which strongly activated units

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suppress their neighbors so that information is routed through a relatively small subset of "winning" units (Desimone, Duncan et al., 1995; Zhou, Friedman and Von Der Heydt, 2000). Existing efficient attention variants partially address computational cost through architectural tricks or token pruning, but they rarely tie sparsity to a competitive selection mechanism and are even less often evaluated in terms of brain-model alignment (Rao, Zhao, Liu, Lu, Zhou and Hsieh, 2021; Beltagy, Peters and Cohan, 2020; Kitaev, Kaiser and Levskaya, 2020).

In this work, we introduce a neuro-inspired split-and-fuse video transformer that explicitly decomposes video processing into two parallel streams and incorporates sparse competitive token selection. The first stream ("what") operates at high spatial resolution and lower frame rate, prioritizing fine-grained object and texture information. The second stream ("where") operates at lower spatial resolution and higher frame rate, emphasizing motion and spatial dynamics. The two streams are processed by separate transformer encoders and fused before classification, either through simple feature concatenation or lightweight cross-attention. This design is motivated by the ventral/dorsal distinction in biological vision and is intended to test whether such pathway specialization can improve both engineering performance and alignment with human neural responses.

To further reduce computational cost and introduce a biologically motivated inductive bias, we replace dense self-attention with a sparse winner-takes-all selection module in each transformer block. Instead of forming a full attention map and computing a weighted sum over all value tokens, the module selects a single most informative token per query position and forwards only that token's representation. During training, this discrete routing is made learnable via a Gumbel-softmax-based (Jang, Gu and Poole, 2017) approximation, while at inference time it reduces to a simple argmax-and-gather operation. This mechanism enforces sparse competition in token space, substantially lowering component memory and compute while preserving representational capacity, and offers a conceptual link to competitive routing in biological circuits (Rao et al., 2021; Desimone et al., 1995).

We evaluate the proposed architecture along three axes. First, we assess video recognition performance and efficiency on Kinetics-400 and Something-Something V2, comparing our split-and-fuse models, with and without sparse selection, against strong convolutional and transformer-based baselines of comparable scale and pretraining (Carreira and Zisserman, 2017; Goyal, Ebrahimi Kahou, Michalski, Materzynska, Westphal, Kim, Haenel, Fruend, Yianilos, Mueller-Freitag et al., 2017; Bertasius et al., 2021). Second, we study robustness to input variation by applying spatial perturbations such as color jitter, geometric transformations, and brightness changes at test time and measuring the resulting accuracy degradation (Fadaei and Dehaqani, 2024). Third, to quantify brain-model alignment, we perform representational similarity analysis between internal model embeddings and time-resolved EEG recordings collected while human participants view naturalistic video clips (Kriegeskorte, Mur and Bandettini, 2008; Schirrmeyer, Springenberg, Fiederer, Glasstetter, Eggenberger, Tangermann, Hutter, Burgard and Ball, 2017). This allows us to ask not only whether the model aligns with aggregate neural responses, but also how its different streams and layers relate to activity over distinct scalp regions (Cichy, Pantazis and Oliva, 2014; Schirrmeyer et al., 2017).

Our results show that the split-and-fuse architecture with sparse selection achieves competitive or state-of-the-art accuracy among models of comparable scale and pretraining while operating on the Pareto frontier of accuracy versus inference time, and that multi-stream variants are more resilient to spatial perturbations than a single-stream TimeSformer baseline (Bertasius et al., 2021). Moreover, the EEG-based analyses reveal modest but consistent improvements in brain-model correspondence, with the full model and its sparse variants exhibiting higher representational similarity to human EEG than monolithic video transformers. The spatial organization of these effects further suggests complementary roles for the "what" and "where" streams, in a manner broadly consistent with ventral and dorsal pathway specializations (Goodale and Milner, 1992; Zareh et al., 2024). Together, these findings support the view that pathway specialization and sparse competition are promising inductive biases for efficient video transformers that better align with human visual processing.

In summary, this work makes three contributions: (i) a neuro-inspired split-and-fuse video transformer with functionally specialized what/where streams, (ii) a sparse winner-takes-all token selection module that serves as an efficient alternative to dense self-attention, and (iii) a systematic EEG-based RSA evaluation of video transformer variants, including stream-wise and regional analyses of brain-model alignment.

2. Related work

2.1. Video action recognition and video transformers

Early deep models for video understanding were predominantly convolutional, extending 2D CNNs to the temporal domain via 3D convolutions or two-stream designs that separate RGB and optical flow inputs (Simonyan

and Zisserman, 2014). Examples include I3D (Carreira and Zisserman, 2017), two-stream CNNs (Simonyan and Zisserman, 2014), and later channel-separated convolutional networks, which demonstrated strong performance on datasets such as Kinetics-400 (Carreira and Zisserman, 2017) and Something-Something V2 (Goyal et al., 2017) but relied on hand-crafted motion inputs or computationally expensive spatiotemporal convolutions. More recent architectures such as SlowFast explicitly introduce dual pathways operating at different frame rates to capture slow semantic content and fast motion, establishing the value of multi-path temporal processing within a CNN framework (Feichtenhofer, Fan, Malik and He, 2019).

Transformer-based models brought a different inductive bias to video recognition by treating frames as sequences of tokens and applying self-attention across space and time (Arnab et al., 2021). TimeSFormer, ViViT, Video Swin, MViT, MTV, and VideoMAE extend image transformers to video using various factorized or hierarchical attention patterns, achieving strong recognition accuracy on large-scale benchmarks (Bertasius et al., 2021; Arnab et al., 2021; Liu et al., 2022b; Li, Wu, Fan, Mangalam, Xiong, Malik and Feichtenhofer, 2022; Yan, Xiong, Arnab, Lu, Zhang, Sun and Schmid, 2022; Wang, Huang, Zhao, Tong, He, Wang, Wang and Qiao, 2023). MTV and related multiview or multiscale transformers process videos through several views at different spatial and temporal scales, but each encoder still uses dense self-attention, so computational cost scales quadratically with the number of tokens (Yan et al., 2022; Fan, Xiong, Mangalam, Li, Yan, Malik and Feichtenhofer, 2021).

More recent video transformers have scaled dramatically through massive web-scale pretraining on billions of unlabeled clips (or millions of hours), enabling foundational models with strong zero-shot transfer to diverse tasks. Models like VideoPrism, InternVideo2, Video Swin Transformer V2, UniFormerV2, and V-JEPA 2 leverage self-supervised learning on datasets exceeding 36M videos or 1M+ hours, achieving state-of-the-art results on Kinetics-400 and Something-Something V2 while supporting downstream applications from action recognition to robotics planning and scientific video analysis (Wang, Li, Li, Yu, He, Chen, Pei, Zheng, Wang, Shi et al., 2024; Zhao, Gundavarapu, Yuan, Zhou, Yan, Sun, Friedman, Qian, Weyand, Zhao et al., 2024; Liu, Hu, Lin, Yao, Xie, Wei, Ning, Cao, Zhang, Dong et al., 2022a; Lu, Jian, Poppe and Salah, 2024; Assran, Bardes, Fan, Garrido, Howes, Muckley, Rizvi, Roberts, Sinha, Zholus et al., 2025). These advances emphasize joint spatiotemporal encoders pretrained via masked modeling, contrastive objectives, or predictive architectures, often with giant ViT backbones (e.g., ViT-g) (Dosovitskiy, Beyer, Kolesnikov, Weissenborn, Zhai, Unterthiner, Dehghani, Minderer, Heigold, Gelly et al.), but still grapple with quadratic attention costs in long-sequence regimes-paving the way for sparsity and pathway innovations like ours.

2.2. Sparse attention mechanisms

Dense self-attention is flexible but expensive, and many variants have been proposed to improve its efficiency (Beltagy et al., 2020; Kitaev et al., 2020; Choromanski, Likhoshesterov, Dohan, Song, Gane, Sarlos, Hawkins, Davis, Mohiuddin, Kaiser et al.). Sparse Transformer (Child, Gray, Radford and Sutskever, 2019), Longformer (Beltagy et al., 2020), Reformer (Kitaev et al., 2020), BigBird (Zaheer, Guruganesh, Dubey, Ainslie, Alberti, Ontanon, Pham, Ravula, Wang, Yang et al., 2020) and related architectures restrict attention patterns using local windows, hashing, or low-rank approximations (Wang, Li, Khabsa, Fang and Ma, 2020); other methods perform dynamic token sparsification, patch shifting, or windowed attention to reduce the number of active tokens, particularly in vision transformers (Rao et al., 2021; Liu, Lin, Cao, Hu, Wei, Zhang, Lin and Guo, 2021; Ryoo, Piergiovanni, Arnab, Dehghani and Angelova, 2021). In video models, temporal shift modules, token shift transformers, and grouped spatiotemporal shift operations approximate temporal interactions with minimal additional parameters, enabling efficient action recognition (Lin, Gan and Han, 2019; Fan, Buch, Wang, Cao, Zhu, Niebles and Fei-Fei, 2020; Zhang, Hao and Ngo, 2021; Li, Shi, Zhang, Cheung, See, Wang, Qin and Li, 2023). Mixture-of-experts layers represent another form of learned conditional computation, where different tokens or regions are routed to different experts with sparse gates (Shazeer, Mirhoseini, Maziarz, Davis, Le, Hinton and Dean, 2017).

These approaches show that sparsity and conditional computation can significantly reduce cost, but most still compute a (possibly structured) weighted sum over many tokens and are not directly motivated by competitive interactions in neural circuits (Rao et al., 2021; Jang et al., 2017). In contrast, our self-selection block replaces dense self-attention with a winner-takes-all mechanism that routes each query to a single selected value token, implementing sparse competitive routing rather than dense averaging (Desimone et al., 1995; Jang et al., 2017). This yields a much simpler attention map, substantially lowering memory and compute compared to dense attention, while providing a direct abstraction of competitive selection and lateral inhibition phenomena in visual cortex (Desimone et al., 1995; Zhou et al., 2000).

2.3. Brain-model alignment

There is longstanding interest in using deep networks as models of the visual cortex and in designing architectures that better mirror biological organization (Yamins and DiCarlo, 2016). Work on object recognition has shown that goal-driven deep convolutional networks trained on large-scale image datasets can predict neural responses in primate inferior temporal cortex and human ventral visual areas to a notable degree (Yamins and DiCarlo, 2016; Dapello et al., 2023). More recently, studies have systematically examined which inductive biases (e.g., texture vs. shape bias, global vs. local processing, architectural choices) influence brain-model alignment, and have highlighted the importance of the similarity metric itself for assessing alignment (Conwell et al., 2024; Soni, Srivastava, Kording and Khosla, 2024; Schrimpf et al., 2018). Additional research has explored brain-score-style benchmarks, convolutional networks for vision neuroscience, and the use of compact deep networks like EEGNet (Lawhern, Solon, Waytowich, Gordon, Hung and Lance, 2018; Schirrmeyer et al., 2017) for EEG decoding and visualization (Schrimpf et al., 2018; Celeghin et al., 2023; Schirrmeyer et al., 2017).

Representational similarity analysis (RSA) has become a standard tool for comparing neural and model representations across time and space (Kriegeskorte et al., 2008). Prior work has used RSA to relate model activations to fMRI, MEG, and scalp EEG recordings, revealing time-resolved correspondences in object recognition and other tasks (Cichy et al., 2014; Cichy and Kaiser, 2019; Kriegeskorte et al., 2008; Schirrmeyer et al., 2017). For dynamic stimuli, recurrent architectures have been shown to better capture visual cortex dynamics than feedforward models (Kietzmann, Spoerer, Sörensen, Cichy, Hauk and Kriegeskorte, 2019), motivating explicit temporal modeling in brain-aligned video systems. However, most studies focus on static images or object recognition, and relatively few examine video transformers or multi-stream architectures in relation to human EEG, especially with explicit modeling of ventral/dorsal-like streams (Goodale and Milner, 1992; Zareh et al., 2024). Moreover, existing efficient attention or token-sparsification methods are rarely evaluated for their impact on brain-model alignment (Rao et al., 2021).

2.4. Positioning of the present work

Our contribution sits at the intersection of these lines of research. Compared to prior video transformers and multi-stream architectures (Bertasius et al., 2021; Feichtenhofer et al., 2019; Zareh et al., 2024), we combine an explicitly ventral/dorsal-inspired what/where split with a sparse winner-takes-all token selection mechanism, and we evaluate this design on large-scale action recognition benchmarks with matched capacity baselines (Carreira and Zisserman, 2017; Goyal et al., 2017). Compared to earlier NeuroAI and RSA-based alignment work (Yamins and DiCarlo, 2016; Kriegeskorte et al., 2008; Schrimpf et al., 2018; Schirrmeyer et al., 2017; Cichy et al., 2014), we provide a systematic EEG evaluation of video transformer variants, including stream-wise and region-wise analyses that probe how pathway specialization and sparse competition affect both overall brain-model similarity and its spatial organization across scalp regions. This allows us to test whether neuro-inspired multi-stream design and competitive token routing serve as useful inductive biases not only for efficiency and robustness, but also for producing representations that are more aligned with human visual dynamics.

3. Methodology

3.1. Model Architecture

The Split-and-Fuse Transformer decomposes a baseline video transformer into two parallel streams that specialize in complementary aspects of the input (overview in Fig. 1) (Goodale and Milner, 1992; Feichtenhofer et al., 2019). The what stream processes frames with high resolution (patch size 16) at a reduced frame rate (8) to prioritize fine-grained spatial structure, whereas the where stream operates on frames with lower resolution (patch size 32) at an increased frame rate (32) to capture motion dynamics.

Each stream consists of an 8-layer transformer encoder derived from a TimeSformer backbone (Bertasius et al., 2021), with separate patch embeddings, positional encodings, and classification tokens. After independent processing, the two final stream embeddings are merged by either (i) feature concatenation, or (ii) a lightweight cross-attention fusion block that performs bidirectional fusion between the two streams.

Let $\mathbf{z}^{\text{what}} \in \mathbb{R}^D$ and $\mathbf{z}^{\text{where}} \in \mathbb{R}^D$ denote the pooled representations of the two streams, and let $\mathbf{W}_{\text{fuse}} \in \mathbb{R}^{D \times 2D}$ be a fusion projection matrix. We define the concatenation fusion as

$$\mathbf{z}^{\text{fuse}} = \mathbf{W}_{\text{fuse}} \begin{bmatrix} \mathbf{z}^{\text{what}} \\ \mathbf{z}^{\text{where}} \end{bmatrix}. \quad (1)$$

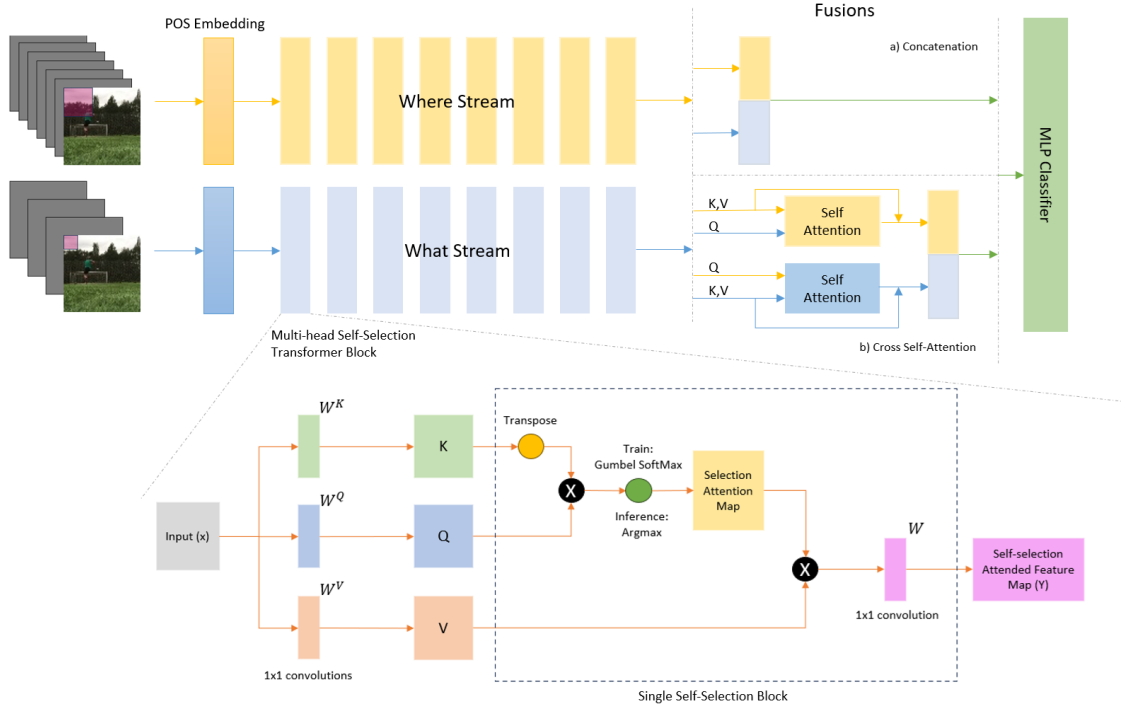


Figure 1: Overview of the neuro-aligned split-and-fuse architecture. The top panel shows the model split into a "what" stream that processes high-resolution frames at a lower frame rate and a "where" stream that processes lower-resolution frames at a higher frame rate to capture motion. The two streams are fused before the classifier via either simple feature concatenation or a cross-attention-based fusion block. The bottom panel illustrates the sparse self-selection module, where a winner-takes-all gate selects the most relevant token for each query, replacing standard dense self-attention.

In the cross-attention fusion block, first, the where stream attends to the what stream (where as queries, what as keys/values), and the resulting representations are added back to the what stream via a residual connection. Then, the what stream attends to the where stream (what as queries, where as keys/values), and the outputs are added back to the where stream. Finally, we concatenate the updated what and where representations before the classifier. Let $\mathbf{Z}^{\text{what}}, \mathbf{Z}^{\text{where}} \in \mathbb{R}^{L \times D}$ denote the token matrices from the what and where streams, respectively, and let $\text{CA}(\cdot, \cdot, \cdot)$ denote a multi-head cross-attention operator that takes queries, keys, and values. We define bidirectional cross-attention fusion as

$$\tilde{\mathbf{Z}}^{\text{what}} = \mathbf{Z}^{\text{what}} + \text{CA}(\mathbf{Z}^{\text{where}}, \mathbf{Z}^{\text{what}}, \mathbf{Z}^{\text{what}}), \quad (2)$$

$$\tilde{\mathbf{Z}}^{\text{where}} = \mathbf{Z}^{\text{where}} + \text{CA}(\mathbf{Z}^{\text{what}}, \mathbf{Z}^{\text{where}}, \mathbf{Z}^{\text{where}}), \quad (3)$$

and the final fused representation as

$$\mathbf{Z}^{\text{fuse}} = \begin{bmatrix} \tilde{\mathbf{Z}}^{\text{what}} \\ \tilde{\mathbf{Z}}^{\text{where}} \end{bmatrix}. \quad (4)$$

3.2. Self-Selection Block

We propose a "winner-takes-all" modification to the canonical self-attention block, replacing the dense softmax weighting over values with a sparse, single-winner selection per query. Inspired by shift-based approximations in convolutional networks (Lin et al., 2019) and by competitive dynamics in biological vision (Desimone et al., 1995), our block computes the usual similarity scores between queries $\mathbf{Q} \in \mathbb{R}^{B \times I \times D}$ and keys $\mathbf{K} \in \mathbb{R}^{B \times J \times D}$, but instead of

passing these through a softmax to produce a full attention map, we select the maximal key for each query channel and use that one value from $\mathbf{V} \in \mathbb{R}^{B \times J \times D}$. In these notations, B denotes the batch size, I is the number of query positions, J is the number of key/value positions, and D denotes the number of dimensions in each query, key, or value vector, that is, the size of the feature-embedding space. During training, we approximate the non-differentiable argmax by a hard Gumbel-softmax (Jang et al., 2017; Herrmann, Bowen and Zabih, 2020) with temperature τ , so that gradients still flow. At inference, we drop back to the exact argmax and a simple gather operation, as illustrated in Fig. 1.

Given query, key, and value tensors

$$\mathbf{Q} \in \mathbb{R}^{B \times I \times D}, \quad \mathbf{K} \in \mathbb{R}^{B \times J \times D}, \quad \mathbf{V} \in \mathbb{R}^{B \times J \times D}, \quad (5)$$

we first form the raw similarity scores

$$\mathbf{S}_{b,i,j} = \mathbf{Q}_{b,i,:} \cdot \mathbf{K}_{b,j,:}, \quad (6)$$

where $\mathbf{S} \in \mathbb{R}^{B \times I \times J}$.

In the standard transformer block (Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin, 2017), one would set

$$\mathbf{A} = \text{softmax}\left(\frac{\mathbf{S}}{\sqrt{d}}\right), \quad \mathbf{O} = \mathbf{A}\mathbf{V}, \quad (7)$$

with $\mathbf{A} \in \mathbb{R}^{B \times I \times J}$.

Instead, we define a selection map $\mathbf{M} \in \{0, 1\}^{B \times I \times J}$ that for each query index (b, i) places a single 1 at the position of the maximal score,

$$\mathbf{M}_{b,i,j} = \begin{cases} 1, & j = \text{argmax}_{j'} \mathbf{S}_{b,i,j'}, \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

and the block output is then computed by a gather:

$$\mathbf{O}_{b,i,:} = \sum_{j=1}^J \mathbf{M}_{b,i,j} \mathbf{V}_{b,j,:} = \mathbf{V}_{b, \text{argmax}_j \mathbf{S}_{b,i,j}, :}. \quad (9)$$

This single-winner approach removes the matrix multiplications that follow a dense attention map, reducing the memory footprint from $O(BIJ)$ to $O(BI)$ and lowering the compute from $O(BIJD)$ to $O(BID)$ in the gather step. However, a direct argmax is non-differentiable, preventing gradient flow. To address this, during training, we employ the Gumbel-softmax trick with hard sampling (Jang et al., 2017):

$$\mathbf{G}_{b,i,j}(\tau) = \frac{\exp((\mathbf{S}_{b,i,j} + \mathbf{g}_{b,i,j})/\tau)}{\sum_{j'} \exp((\mathbf{S}_{b,i,j'} + \mathbf{g}_{b,i,j'})/\tau)}, \quad (10)$$

where $\mathbf{g}_{b,i,j} \sim \text{Gumbel}(0, 1)$ are fresh noise samples, and $\tau > 0$ is a temperature parameter. During each training forward pass, we assign

$$\mathbf{A}_{b,i,j} = \text{onehot}(\text{argmax}_j \mathbf{G}_{b,i,j}), \quad (11)$$

and in backprop, we use

$$\frac{\partial \mathbf{A}}{\partial \mathbf{S}} = \frac{\partial \mathbf{G}}{\partial \mathbf{S}}. \quad (12)$$

This hard Gumbel-softmax ensures that the forward output matches exactly the discrete selection map $\mathbf{M}$, while the backward pass uses the continuous softmax gradient of $\mathbf{G}$. By annealing τ (for example, from 1.0 down toward

0.1), the distribution of $\mathbf{G}$ increasingly concentrates on the actual maximum, thus matching the argmax more closely. We can compute

$$\lim_{\tau \rightarrow 0^+} \mathbf{G}_{b,i,j}(\tau) = \begin{cases} 1, & j = \operatorname{argmax}_{j'} (\mathbf{S}_{b,i,j'} + \mathbf{g}_{b,i,j'}), \\ 0, & \text{otherwise.} \end{cases} \quad (13)$$

During training, the term $\mathbf{g}_{b,i,j} \sim \text{Gumbel}(0, 1)$ introduces random noise to every score. At inference, we drop the noise and compute the selection mask $\mathbf{M}$ by applying argmax directly, then use this mask to gather the corresponding value vectors as in Eq. (9). This yields deterministic routing at test time while retaining nonzero gradients for $\mathbf{S}$ through the continuous relaxation during training.

4. Experiments and results

4.1. Experimental setup

All models are implemented in PyTorch and trained using mini-batches of fixed-length clips on NVIDIA RTX 4060 Ti GPUs. We consider different variants of our architecture: the what-only stream, the where-only stream, the "S What" stream which is the what-only stream with self-selection, the "S Where" stream which is the where-only stream with self-selection, the dual-stream WW-Former (without self-selection), and the dual-stream SWW-Former (with self-selection blocks in all transformer layers), each instantiated with either concatenation fusion or bidirectional cross-attention fusion. To stabilize sparse selection in the related models, the Gumbel-softmax temperature τ (Jang et al., 2017) is initialized at a moderate value and annealed exponentially over training epochs, sharpening the winner-takes-all distribution.

For video recognition, we train and evaluate on Kinetics-400 (Carreira and Zisserman, 2017) and Something-Something V2 (SSv2) (Goyal et al., 2017). On Kinetics-400, all models are initialized from ImageNet-pretrained weights (Deng, Dong, Socher, Li, Li and Fei-Fei, 2009). For SSv2, we initialize from the corresponding Kinetics-400 checkpoints and fine-tune. In both datasets, videos are decoded into fixed-length clips, uniformly sampled over time, resized to a target spatial resolution, and normalized channel-wise; the what stream receives high-resolution, low-frame-rate clips, while the where stream receives low-resolution, high-frame-rate clips generated from the same raw video. We use identical data pipelines and augmentation strategies (including random cropping, horizontal flipping, and standard color jitter) across all models and baselines (including TimeSFormer, VideoSwin, MViT, MTV, and VideoMAE) to ensure a fair comparison. Classification performance is reported as top-1 and top-5 accuracy on the held-out validation set for each dataset.

We train all models with the AdamW optimizer, using a maximum learning rate of 3×10^{-4} , one cycle learning-rate decay, and a weight decay of 0.005. Each model is trained for 30 epochs on Kinetics-400 and 20 additional epochs on SSv2 with a global batch size of 64 clips (8 clips per GPU step on our NVIDIA RTX 4060 Ti setup). Input clips consist of 8 frames for the what stream and 32 frames for the where stream, sampled uniformly from each video. We use label smoothing with factor 0.05 and standard dropout in the transformer blocks; all hyperparameters are kept fixed across our model variants to isolate the effect of architectural changes. Table 1 summarizes the key setup details for all evaluated models, including parameter counts, input resolutions, and release years of baselines.

4.2. Video recognition performance

To assess the impact of the split-and-fuse design and the sparse self-selection blocks on recognition performance, we evaluate four main variants of our architecture: WW-Former and SWW-Former, each instantiated with either concatenation or bidirectional cross-attention fusion. For each variant, we also report results for the individual what-only and where-only streams. These models are compared against the original TimeSformer backbone (Bertasius et al., 2021) and strong representative video architectures, including MTV (Yan et al., 2022), VideoMAE (Wang et al., 2023), MViT (Fan et al., 2021; Li et al., 2022), Video Swin (Liu et al., 2022b,a), and SlowFast (Feichtenhofer et al., 2019), on Kinetics-400 and SSv2 (Carreira and Zisserman, 2017; Goyal et al., 2017); the complete top-1 and top-5 accuracies are summarized in Table 2.

As reported in Table 2, the best-performing variant, SWW-Former with attention-based fusion, achieves 82.55% and 73.18% top-1 accuracy on Kinetics-400 and SSv2, respectively, outperforming similar-scale state-of-the-art models by more than 0.75 percentage points and 1 percentage point, excluding models with additional web-scale pretraining.

Table 1

Setup details for baseline and proposed models. The input size indicates frame count $\times$ spatial resolution (e.g., $8 \times 224 \times 224$ for 8-frame, 224×224 clips). For multi-stream models, largest frame count and spatial resolution is reported. Our models are derived from a 2021 TimeSformer backbone (Bertasius et al., 2021) and augmented with self-selection where indicated.

Model	Release Year	# Params	Input Size
I3D-R50 (Carreira and Zisserman, 2017)	2017	28.04 M	$8 \times 224 \times 224$
SlowFast-R50 (Feichtenhofer et al., 2019)	2019	34.57 M	$32 \times 224 \times 224$
SlowFast-R101 (Feichtenhofer et al., 2019)	2019	62.83 M	$32 \times 224 \times 224$
CSN-R101 (Tran, Wang, Torresani and Feiszli, 2019)	2019	22.21 M	$32 \times 224 \times 224$
TimeSformer (Bertasius et al., 2021)	2021	121.57 M	$8 \times 224 \times 224$
TimeSformer-HR (Bertasius et al., 2021)	2021	122.02 M	$16 \times 448 \times 448$
TimeSformer-L (Bertasius et al., 2021)	2021	121.61 M	$64 \times 224 \times 224$
VideoSwin-B (V1) (Liu et al., 2022b)	2021	88.05 M	$32 \times 224 \times 224$
MViT-B (V2) (Li et al., 2022)	2022	36.61 M	$32 \times 224 \times 224$
MTV-B (Yan et al., 2022)	2022	310 M	$32 \times 224 \times 224$
VideoMAE (V2) (Wang et al., 2023)	2023	86.53 M	$16 \times 224 \times 224$
VideoSwin-G (V2) (Liu et al., 2022a)	2022	3 B	$8 \times 320 \times 320$
VideoPrism-G (Zhao et al., 2024)	2024	1 B	$8 \times 288 \times 288$
InternVideo (V2) (Wang et al., 2024)	2024	6 B	$16 \times 224 \times 224$
FTP-UniFormer (V2) (Lu et al., 2024)	2024	620 M	$32 \times 224 \times 224$
V-JEPA-G (V2) (Assran et al., 2025)	2025	1 B	$16 \times 384 \times 384$
What Stream	2021	57.75 M	$8 \times 224 \times 224$
S What Stream	2026	57.75 M	$8 \times 224 \times 224$
Where Stream	2021	83.07 M	$32 \times 224 \times 224$
S Where Stream	2026	83.07 M	$32 \times 224 \times 224$
WW-Former Concat	2026	141.44 M	$32 \times 224 \times 224$
SWW-Former Concat	2026	141.44 M	$32 \times 224 \times 224$
WW-Former Attention	2026	146.17 M	$32 \times 224 \times 224$
SWW-Former Attention	2026	146.17 M	$32 \times 224 \times 224$

4.3. Computational efficiency

Table 3 reports average inference time, GFLOPs, and peak GPU memory usage for our dual-stream models and comparable transformer baselines, measured for a batch size of 8 on an RTX 4060 Ti and averaged over 100 runs.

Self-selection blocks yield consistent efficiency gains across both fusion strategies. For attention-fusion variants, WW-Former requires 444.94 ms per batch, 1763.90 GFLOPs, and 3.34 GB peak memory, whereas SWW-Former with self-selection reduces these by 4.7% to 423.83 ms, 3.0% to 1710.33 GFLOPs, and 2.1% to 3.27 GB while boosting top-1 accuracy (80.16% to 82.55% on Kinetics-400; 69.46% to 73.18% on SSv2; see Tables 1 and 2). A similar pattern holds for concatenation-fusion variants. In the accuracy-latency plane (Fig. 2 and Fig. 3), SWW-Former variants occupy the Pareto frontier among models of comparable-scale and pretraining (no other databases for pretraining besides ImageNet (Deng et al., 2009) and Kinetics-400 (Carreira and Zisserman, 2017)).

While concatenation fusion minimizes additional compute (141.44M params, 423.43 ms inference for SWW-Former Concat), cross-attention fusion adds modest overhead (146.17M params, +0.4 ms) but remains efficient due to lightweight bidirectional design. Both strategies benefit from self-selection sparsity, with attention fusion preserving Pareto dominance despite extra parameters.

4.4. Gumbel-softmax temperature ablation

The temperature parameter τ in the hard Gumbel-softmax (Jang et al., 2017) controls how sharply the selection distribution concentrates on the highest-scoring key for each query in the self-selection block. During training, larger τ yields a softer, more exploratory distribution over keys, while smaller τ makes the sampling behave more like a hard argmax, approximating the deterministic inference-time behavior. In the main experiments, τ is annealed from a moderate initial value toward a lower final value, allowing early training to benefit from smoother gradients while

Table 2

Comparison with previous best results on video action classification datasets. Acc@N stands for Top N Accuracy. The models listed in the middle section were reported to be pre-trained on large-scale web data (other than ImageNet (Deng et al., 2009) and Kinetics-400 (Carreira and Zisserman, 2017) datasets), and their results are not directly comparable to other models which are listed.

Model	# Params	Kinetics-400		SS-V2	
		Acc@1	Acc@5	Acc@1	Acc@5
I3D-R50 (Carreira and Zisserman, 2017)	28.04 M	73.27	90.7	62.84	-
SlowFast-R50 (Feichtenhofer et al., 2019)	34.57 M	76.94	92.69	61.68	86.92
SlowFast-R101 (Feichtenhofer et al., 2019)	62.83 M	77.9	93.27	63	-
CSN-R101 (Tran et al., 2019)	22.21 M	77	92.9	-	-
TimeSFormer (Bertasius et al., 2021)	121.57 M	77.9	93.2	59.1	85.6
TimeSFormer-HR (Bertasius et al., 2021)	122.02 M	79.6	94	61.8	86.9
TimeSFormer-L (Bertasius et al., 2021)	121.61 M	80.6	94.7	62	87.5
VideoSwin-B (V1) (Liu et al., 2022b)	88.05 M	82.7	95.5	69.6	92.7
MViT-B (V2) (Li et al., 2022)	36.61 M	80.3	94.69	72.1	-
MTV-B (Yan et al., 2022)	310 M	81.8	95	67.6	90.1
VideoMAE (V2) (Wang et al., 2023)	86.53 M	81.5	95.1	71.2	92
VideoSwin-G (V2) (Liu et al., 2022a)	3 B	86.8	-	-	-
VideoPrism-G (Zhao et al., 2024)	1 B	87.2	-	68.5	-
InternVideo-6B (V2) (Wang et al., 2024)	6 B	92.1	-	77.4	-
FTP-UniFormer (V2) (Lu et al., 2024)	620 M	93.4	99.3	79.8	98.9
V-JEPA-G (V2) (Assran et al., 2025)]	1 B	87.3	-	77.3	-
What Stream	57.75 M	72.31	90.01	55.61	85.18
S What Stream	57.75 M	77.48	93.54	58.48	86.34
Where Stream	83.07 M	72.46	90.61	58.45	86.01
S Where Stream	83.07 M	78.12	94.4	60.67	86.9
WW-Former Concat	141.44 M	78.91	93.86	68.03	90.92
SWW-Former Concat	141.44 M	81.5	95.58	69.93	92.31
WW-Former Attention	146.17 M	80.16	94.56	69.46	91.99
SWW-Former Attention	146.17 M	82.55	96.01	73.18	93.65

late training encourages commitment to single winners, similar in spirit to prior work that uses Gumbel-softmax for differentiable channel selection (Herrmann et al., 2020).

To assess the sensitivity of our model to the choice of Gumbel-softmax temperature τ , we trained SWW-Former variants with fixed τ values ranging from 0.1 to 10 and measured mean top-1 accuracy and standard deviation over 5 runs on the Kinetics-400 validation set.

As shown in Fig. 4, performance is relatively stable across the tested range. As a result all temperatures achieve accuracy within roughly two percentage points of each other, indicating that the self-selection mechanism is robust to the exact choice of τ . The highest mean accuracy is observed around $\tau = 0.3$, with $\tau = 0.1$ and larger values (e.g., $\tau \geq 5$) yielding slightly lower but still comparable performance. These results suggest that using a moderately low temperature provides a good practical default without requiring fine-tuning of τ .

4.5. Invariance and robustness analysis

To evaluate generalization beyond clean validation data, we assessed model robustness by applying random spatial and photometric perturbations at test time, following prior work on invariances and input-variance robustness (Tschanen, Djolonga, Ritter, Mahendran, Houlsby, Gelly and Lucic, 2020; Fadaei and Dehaqani, 2024), measuring the relative drop in top-1 accuracy averaged over 5 trials on the Kinetics-400 (Carreira and Zisserman, 2017) and Something-Something V2 (Goyal et al., 2017) validation sets. Specifically, we applied random spatial augmentations including color jittering, flipping, reversing, rotation, cropping, and adjustments to brightness, contrast, and hue. Detailed application of these augmentations are included in our shared codebase.

Table 3

Inference efficiency comparison (batch size 8 on RTX 4060 Ti, averaged over 100 runs). Inference time is in ms per batch; GFLOPs and peak CUDA memory (GB) are also reported. Self-selection variants (S) consistently reduce compute and latency relative to their dense-attention counterparts while improving accuracy.

Model	Avg. Inference Time	GFLOPs	Peak CUDA Memory
I3D-R50 (Carreira and Zisserman, 2017)	84.61 ms	229.35	789.11 MB
SlowFast-R50 (Feichtenhofer et al., 2019)	274.78 ms	406.84	1.06 GB
SlowFast-R101 (Feichtenhofer et al., 2019)	361.05 ms	777.51	1.17 GB
CSN-R101 (Tran et al., 2019)	293.46 ms	460.89	2.79 GB
TimeSFormer (Bertasius et al., 2021)	386.73 ms	1520.44	3.1 GB
TimeSFormer-HR (Bertasius et al., 2021)	3932.16 ms	2149.44	9.08 GB
TimeSFormer-L (Bertasius et al., 2021)	3129.79 ms	2160.33	5.96 GB
VideoSwin-B (V1) (Liu et al., 2022b)	1241.34 ms	1310.9	6.98 GB
MViT-B (V2) (Li et al., 2022)	1147.91 ms	1216.16	5.8 GB
MTV-B (Yan et al., 2022)	2683.86 ms	1453.01	5.98 GB
VideoMAE (V2) (Wang et al., 2023)	446.54 ms	814.79	2.43 GB
What Stream	178.74 ms	721.93	1.6 GB
S What Stream	170.25 ms	716.55	1.52 GB
Where Stream	266.35 ms	1041.93	2.07 GB
S Where Stream	255.63 ms	1020.15	2.01 GB
WW-Former Concat	444.8 ms	1763.86	3.33 GB
SWW-Former Concat	423.43 ms	1709.55	3.24 GB
WW-Former Attention	444.94 ms	1763.9	3.34 GB
SWW-Former Attention	423.83 ms	1710.33	3.27 GB

Table 4

Reported top-1 accuracy and the calculated accuracy drop without and with augmentations in Kinetics-400 (Carreira and Zisserman, 2017) and Something-Something V2 (Goyal et al., 2017) Datasets.

Model	Kinetics-400			SS-V2		
	Normal	Augmented	Drop	Normal	Augmented	Drop
TimeSFormer (Bertasius et al., 2021)	77.9	76.61	1.29	59.1	38.09	21.01
What Stream	72.31	71.17	1.14	55.61	36.09	19.52
S What Stream	77.48	76.44	1.04	58.48	40.1	18.38
Where Stream	72.46	71.51	0.95	58.45	40.05	18.4
S Where Stream	78.12	77.11	1.01	60.67	44.87	15.8
WW-Former Concat	78.91	78.27	0.64	68.03	52.03	16
SWW-Former Concat	81.5	80.83	0.67	69.93	55.83	14.1
WW-Former Attention	80.16	79.4	0.76	69.46	53.14	16.32
SWW-Former Attention	82.55	81.72	0.83	73.18	58.49	14.69

Table 4 and Fig. 5 show the resulting drop in accuracy. Our proposed multi-stream models are more resilient to these changes, with a smaller drop in accuracy compared to the TimeSFormer baseline (p-value < 0.01; bootstrap tests with N=100000). Larger accuracy drops on SSv2 (~14-21%) versus Kinetics (~0.6-1.3%) reflect dataset differences. SSv2 emphasizes fine-grained temporal ordering and human-object interactions sensitive to spatial augmentations like rotation/cropping, while Kinetics-400 focuses on broader action recognition more invariant to such perturbations.

4.6. EEG representational similarity analysis

To assess whether our split-and-fuse transformers mirror human brain dynamics, and following prior model-brain alignment work (Yamins and DiCarlo, 2016; Schrimpf et al., 2018; Conwell et al., 2024; Celeghin et al., 2023), we recorded 128-channel scalp EEG (g.HIamp/g.GAMMASys, 10-10 layout plus interleaved) at 512 Hz (downsampled to 200 Hz) while participants viewed selected clips from the HVU dataset (Diba, Fayyaz, Sharma, Paluri, Gall,

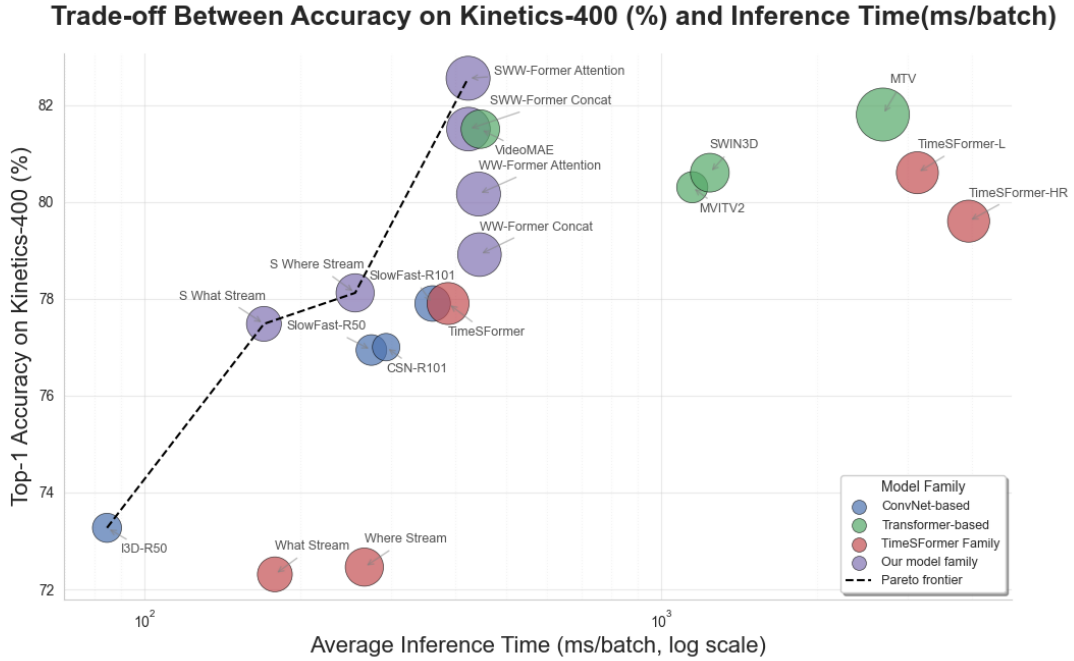


Figure 2: Trade-off between top-1 accuracy and inference time (ms per batch) on Kinetics-400 (Carreira and Zisserman, 2017) for our models and others of similar size and pretraining. Models in the top-left region are both more accurate and faster. Bubble size indicates the number of trainable parameters.

Stiefelhagen and Gool, 2020). Signals were high-pass filtered (1 Hz), notch-filtered (49-51 Hz), re-referenced to the channel average, epoched (-200 ms to 3.4 s post-onset; -200 ms to 400 ms for images), baseline-corrected, and cleaned of artifacts via visual inspection and ICA.

Thirty healthy adults (16 M/14 F, ages 20 to 30) from a pre-screened pool gave written informed consent under an IRB-approved protocol compliant with the Code of Ethics. Exclusion criteria included neurological disorders, uncorrected vision deficits, epilepsy, and head injury; a general practitioner verified eligibility. We handled the personal data in accordance with institutional policy. Each participant was compensated over the national minimum wage to cover time and travel expenses. We included the participant instructions, task screenshots, and consent form in the appendix B. The EEG data can be made available upon reasonable request to the corresponding author, but will not be made publicly available due to privacy concerns, institutional restrictions, and other considerations.

We selected 252 intact 3 s clips from eight categories (Animal, Art, Food, Garden, Kid, Music, Sports, and Vehicle) and 21 subcategories from the HVU dataset (Diba et al., 2020); from which we randomly scrambled 84 videos by frame shuffling, and we have shown their middle frame as 50 ms static images as separate stimuli. Stimuli subtended a 5° visual angle, ran at 60 Hz on a gray background, and appeared in three trial types: coherent video, scrambled video, and static image. Each two-hour session comprised 336 video trials and 252 image trials (each repeated four times), with 400 ms inter-trial intervals, pseudo-random fixation-cross color changes for attention checks, and rest breaks every ~15 minutes.

For model comparison, we first applied multi-class linear discriminant analysis (LDA) (Fisher, 1936) at each time point to decode the 21 stimulus subcategories from the EEG, providing a simple baseline alongside more complex EEG decoding approaches (Schirrmeyer et al., 2017). This revealed a transient accuracy peak at approximately 200 ms after video onset, indicative of early category-selective visual processing consistent with prior MEG/EEG studies (Cichy et al., 2014)(Fig. 6 a).

To quantify the alignment between model and human EEG representations, we performed time-resolved Representational Similarity Analysis (RSA) (Kriegeskorte et al., 2008). For each model, the pairwise dissimilarities of its video embeddings were computed and flattened into Representational Dissimilarity Matrices (RDMs). EEG RDMs

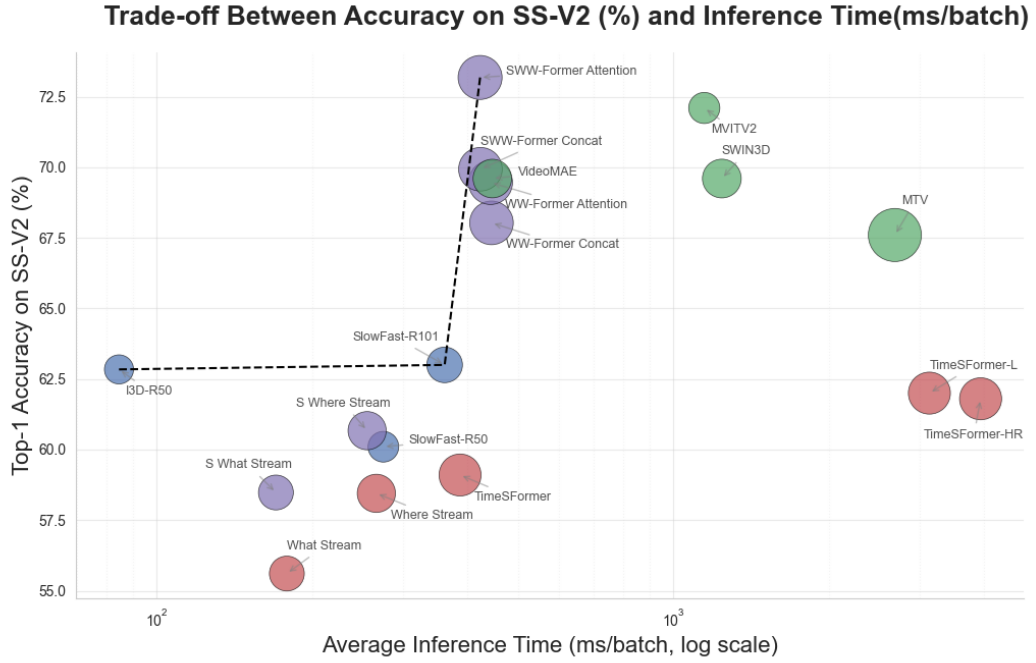


Figure 3: Trade-off between top-1 accuracy and inference time (ms per batch) on SSv2 (Goyal et al., 2017) for our models and others of similar size and pretraining. Models in the top-left region are both more accurate and faster. Bubble size indicates the number of trainable parameters.

were constructed for each participant at each time point from scalp-recorded responses, and flattened analogously. Spearman correlation coefficients were then computed between the model and EEG RDMs at each time point, yielding participant-level time courses of representational similarity. To contextualize these correlations, we estimated a noise ceiling reflecting the maximum explainable variance given inter-subject reliability. Specifically, for each time point, the upper bound was calculated as the mean Spearman correlation between each participant's EEG RDM and the group-average RDM including that participant. All correlations were Fisher-Z transformed before averaging and smoothed with a five-timepoint moving average to reduce noise. The resulting noise ceiling provides a reference for the best achievable model-EEG alignment in our dataset, enabling normalization of model correlations as a fraction of explainable variance.

This yielded brain-model correspondence time courses for each model, all peaking around 195 ms and mirroring the LDA profile (Fig. 6 b). To directly compare models, we examined RSA correlations at the 195 ms peak (Fig. 6 c). We conducted two-sided pairwise Wilcoxon signed-rank tests to assess the statistical significance of differences between model pairs (Fig 6 d). Our SWW-Former model with concatenation fusion achieves 0.18 RSA correlation, which, excluding the SWW-Former with attention fusion, is significantly higher than all tested models ($p < 0.05$ two-sided Wilcoxon signed-rank test against all other baselines) and reached up to 78% of the estimated noise ceiling at the peak time, indicating substantial capture of neural representational structure.

The what and where streams show closely similar correlation values of 0.11 and 0.10, respectively (p -value = 0.325; two-sided pairwise Wilcoxon signed-rank test), indicating that both streams contribute similarly to the model and its neuro-alignment. Furthermore, the Concatenation and Attention fusion variants of WW-Former, with multi-stream, and S What and S Where stream models, which implement self-selection blocks instead of self-attention blocks, achieve 0.16, 0.15, 0.15 and 0.14 correlation, respectively. These results show significant improvement over the standard What and Where streams (For every comparisons of these variants with each of What and Where streams, p -value < 0.005 ; two-sided pairwise Wilcoxon signed-rank test), demonstrating that each of our introduced contributions helps with the neural alignment of the model. SWW-Former with concatenation achieves 0.18 correlation comparing to 0.17

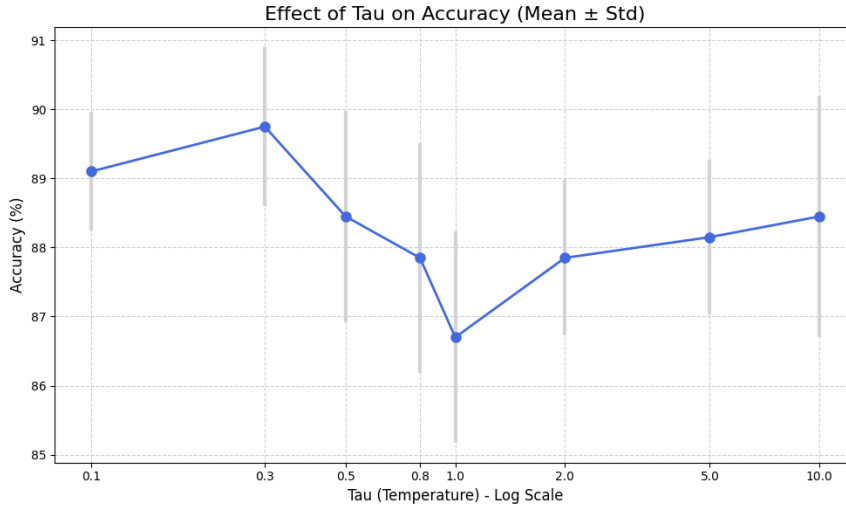
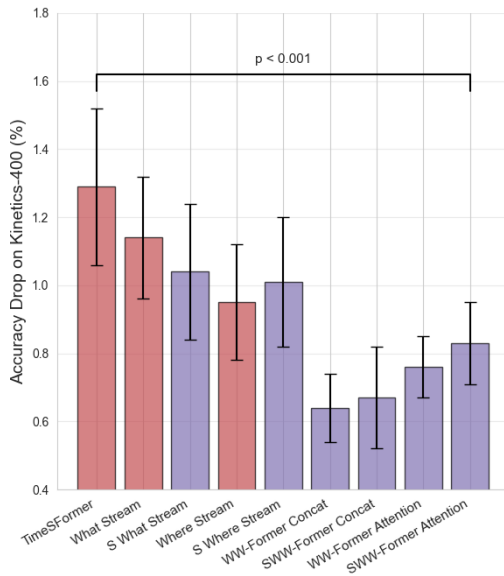


Figure 4: Effect of fixed Gumbel-softmax temperature τ on top-1 validation accuracy of SWW-Former (Attention), with error bars indicating one standard deviation across 5 runs. Accuracy varies by less than about two percentage points across the tested range and peaks around $\tau = 0.3$.

a) Accuracy Drop When Applying Spatial Augmentations on Kinetics-400



b) Accuracy Drop When Applying Spatial Augmentations on SS-V2

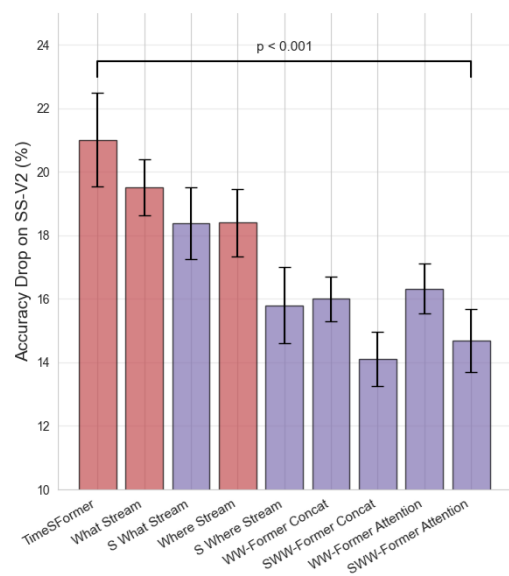


Figure 5: Percentage drop in top-1 accuracy under spatial augmentations on (a) Kinetics-400 (Carreira and Zisserman, 2017) and (b) Something-Something V2 (Goyal et al., 2017). The error bars report the standard deviation of 5 trials and p-values test differences from TimeSFormer and are calculated by bootstrap test (N=100,000).

correlation by SWW-Former attention (P-value >0.33; two-sided paired Wilcoxon signed-rank test) which suggests our different fusion mechanics have no significant effect on brain-model alignment.

Finally, we examined the spatial organization of brain-model correspondence by mapping channel-wise RSA onto the scalp. We computed channel-wise significance tests relative to the best-aligned architecture and aggregated correlations within canonical lobar ROIs. This analysis, summarized in Fig. 7, shows that the SWW-Former variants yield higher, more spatially structured correlations in parietal, lateral temporal, and other regions than monolithic

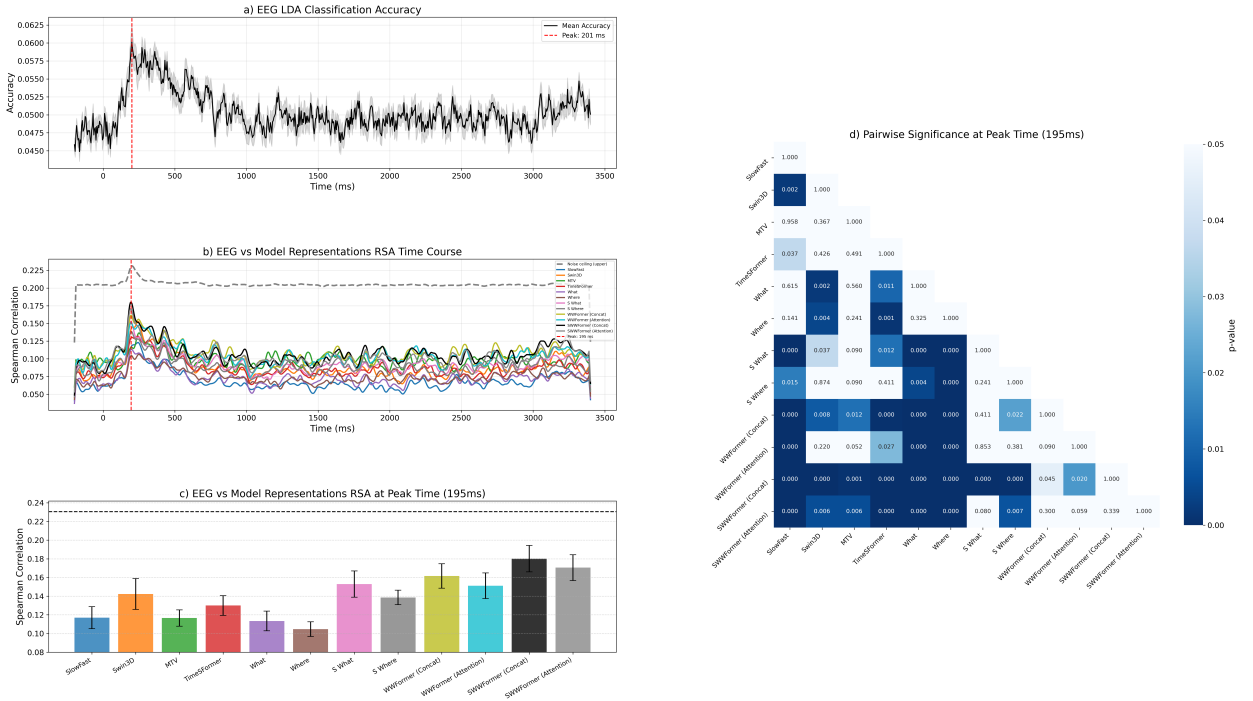


Figure 6: Model-EEG alignment. (a) Time-resolved multiclass LDA decoding accuracy for 21 visual subcategories. (b) Time-resolved mean Spearman RSA correlations between model RDMs and EEG RDMs with noise ceiling. (c) Peak mean RSA correlations (195 ms) across models (The error bars indicate SEM), with noise ceiling. (d) Pairwise Wilcoxon signed-rank tests on peak RSA scores.

baselines. The SWW-Former recorded average correlations of 0.10, 0.14, 0.07, 0.07, and 0.06 for frontal, parietal, temporal, occipital, and central region electrodes, respectively. The two-sided paired t-tests with SWW-Former variants are reported in Fig. 7. This analysis shows that the fused sparse model is consistently among the top performers across various lobes.

Furthermore, Fig. 7 ROI analysis confirms that the what/S What streams align better with temporal electrodes (ventral-like), while Where/S Where streams align better with parietal (dorsal-like), consistent with classic dorsal-ventral dissociations (Goodale and Milner, 1992; Creem and Proffitt, 2001) and recent DVSD modeling (Zareh et al., 2024) ($p < 0.05$, paired t-tests against the counterpart stream for each comparison). We also included layer-wise RSA analysis of the S What and S Where streams in the appendix A, detailing how different layers of our streams correlate with the recorded data.

5. Discussion

This work presented a Neuro-Aligned Split-and-Fuse Transformer. It decomposes video processing into specialized "what" and "where" streams and augments them with a sparse self-selection mechanism. Across Kinetics-400 and Something-Something V2, the split-and-fuse design with self-selection achieved competitive or state-of-the-art performance among models of comparable scale without relying on massive web pre-training. Our model also operates at the Pareto frontier of accuracy versus inference time in both datasets. These architectural changes yielded representations that are more closely correlated with human EEG responses than those from conventional monolithic transformers, suggesting that explicit pathway separation and winner-takes-all selection are beneficial for both engineering and NeuroAI, and contribute to ongoing efforts to align high-level visual representations in brains and machines (Conwell et al., 2024; Celeghin et al., 2023; Soni et al., 2024) and in line with classic dorsal-ventral dissociations and competitive-attention mechanisms in vision (Goodale and Milner, 1992; Creem and Proffitt, 2001; Desimone et al., 1995; Itti and Koch, 2001).

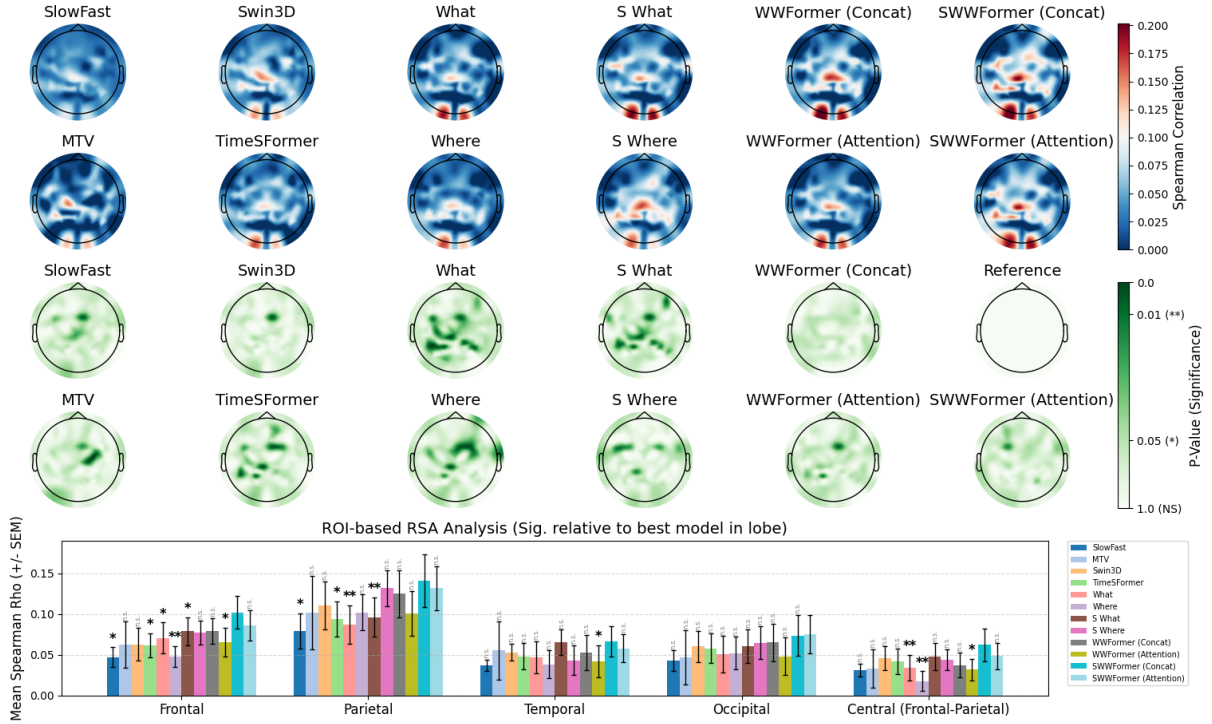


Figure 7: Channel-wise and regional RSA between video models and EEG at peak latency (195 ms). The top two rows show scalp topographies of Spearman correlations. The middle two rows show significance maps from paired t-tests against best model per lobe where darker green indicates lower p-values and significant difference from SWW-Former with concatenation fusion. The plot shows mean RSA by ROI including frontal, parietal, temporal, occipital, central (fronto-parietal) regions, with markers for two-sided paired t-tests against best model per ROI. The error bars in this plot indicate SEM.

Beyond summarizing empirical gains, a central claim of this paper is mechanistic: sparse competitive selection together with pathway specialization constrains the model to forward only a small subset of informative states, echoing sparse and dynamic selection mechanisms in modern neural architectures (Shazeer et al., 2017; Beltagy et al., 2020; Kitaev et al., 2020; Rao et al., 2021), which both reduces computation and produces a representational geometry that more closely resembles early/mid-latency cortical responses reported in human neuroimaging (Cichy et al., 2014; Yamins and DiCarlo, 2016; Schrimpf et al., 2018). Our analyses are consistent with this account in several ways. Replacing dense attention by single-winner selection reduces latency markedly while largely preserving top-line accuracy and robustness (Fig. 2 and Fig. 3). Furthermore, the RSA results (Fig. 6 and Fig. 7) shows that whole and partial brain-model alignments increases in models with self selection.

That said, the neural-alignment results require careful qualification. The peak Spearman correlation we report for the SWW-Former (0.18) is modest in absolute magnitude, reflecting a small but reliable fraction of between-subject variance in scalp EEG. To contextualize this effect, we computed a noise ceiling, representing the best-case model-EEG correlation given inter-subject reliability in our dataset (Kriegeskorte et al., 2008). At the time of peak correlation (195 ms), the SWW-Former explains approximately 78% of the explainable variance, based on the mean noise ceiling (correlation = 0.23). Reporting correlations alongside noise-ceiling fractions makes clear that our models capture neural responses more closely than baseline transformers, while substantial neural variance remains unexplained.

Several alternative explanations could inflate model-brain similarity, and we performed control analyses to address them. Time-shifted RSA (random circular temporal shifts of EEG relative to stimuli) yields correlations near zero, ruling out trivial temporal autocorrelation as the sole driver. Furthermore, we computed RDMs from other convolutional and transformer based models such as SlowFast (Feichtenhofer et al., 2019), Swin3D (Liu et al., 2022b),

MTV (Yan et al., 2022) and TimeSFormer (Bertasius et al., 2021), which produced substantially lower correlations than our models (Fig. 6).

We further examined the spatial organization of model-brain correspondence by mapping channel-wise RSA across the scalp and aggregating correlations within canonical frontal, parietal, temporal, occipital and central regions (Fig. 7). This analysis revealed that our split-and-fuse models produce more structured and generally higher correlations over occipital, parietal and lateral temporal electrodes than monolithic transformer baselines, indicating that sparse multi-stream processing yields representations that better capture the distributed geometry of human visual responses. Moreover, the what and where streams show complementary alignment profiles, with the what stream more strongly correlated in temporal regions and the where stream more strongly correlated in parietal regions, consistent with the functional dissociation of ventral and dorsal visual pathways (Goodale and Milner, 1992; Creem and Proffitt, 2001). Together, these spatially resolved EEG results extend our findings beyond simple decoding accuracy or global similarity scores, and position our framework as a richer, temporally and regionally specific testbed for NeuroAI hypotheses about sparse competition, pathway specialization and their roles in shaping efficient visual representations.

Methodological limitations must be acknowledged. First, scalp EEG provides limited spatial precision and mixes cortical sources; thus channel-wise topographies must be interpreted cautiously. Source-reconstruction or higher-resolution modalities (MEG, fMRI, or intracranial recordings) could more precisely test cortical mapping claims. Second, our sample size ($N = 30$) allows for reliable estimation of group-level RSA correlations. While individual participants' correlations are modest in magnitude (0.1 to 0.18), averaging across participants yields stable time courses, and the time-resolved noise ceiling contextualizes these effects relative to the maximum explainable variance in our dataset. Thus, the observed neural alignment is robust at the group level, even though single-participant correlations remain small. However, larger samples and more diverse stimuli would strengthen claims about generality. Third, the selection mechanism was evaluated only in a TimeSFormer-derived backbone and two fusion strategies; whether comparable gains persist across other transformer families or large-scale pretraining regimes remains to be tested.

These caveats suggest several concrete follow-up steps. One direction is to extend the neuro-alignment analysis to fMRI, MEG, or intracranial recordings and to additional datasets, to separate genuine cortical principles from data-specific regularities. Another is to apply sparse What-Where transformers to temporal localization, video-language grounding, and online decision-making, and ask whether the efficiency gains we observe here actually improve behavior in interactive or resource-constrained settings. A longer-term avenue is to embed self-selection into scalable video backbones and multimodal foundation models, aiming to jointly improve data efficiency, computational cost, and alignment with human perception and cognition.

6. Conclusion

In this work, we introduced neuro-aligned split-and-fuse video transformers with sparse winner-takes-all self-selection, achieving competitive or SOTA accuracy (82.55% on Kinetics-400 (Carreira and Zisserman, 2017), 73.18% on SSv2 (Goyal et al., 2017)), inference Pareto efficiency, robustness, and EEG alignment ($\rho = 0.18$, 78% of the noise ceiling (Kriegeskorte et al., 2008)) superior to baselines. In sum, our results suggest that pathway specialization and sparse competition are promising inductive biases for efficient, neuro-aligned video models. While the gains in brain-model correspondence are modest, they are consistent, spatially structured, and robust to several controls, supporting further exploration of biologically inspired sparsity in transformer architectures and paving the way for NeuroAI in resource-constrained vision tasks.

CRedit authorship contribution statement

Amir Hosein Fadaei: Conceptualization, Data curation, Methodology, Software, Investigation, Formal analysis, Validation, Visualization, Writing - original draft, Writing - review and editing. **Mahyar Maleki:** Data curation, Investigation, Formal analysis. **Mohammad-Reza A. Dehaqani:** Conceptualization, Funding acquisition, Project administration, Supervision, Validation, Writing - review and editing.

7. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used AI-based language editing tools solely to improve the grammar and readability of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

8. Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

9. Data and Code availability

The video datasets used in this study, including Kinetics-400, Something-Something V2 and the HVU dataset, are available from the previously published works cited in this manuscript. EEG data collected for this study are not publicly available due to institutional regulations and participant privacy considerations, but are available from the corresponding author upon reasonable request.

The research codebase, trained models, and all related codes for various research and tests conducted in this manuscript are available in our public github repository: <https://github.com/amfad33/SWW-Former>

10. Acknowledgement

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A. Appendix: Layer-wise RSA for the Streams

The layer-wise RSA analyses for the S What and S Where streams (Fig. 8 and Fig. 9) show that neural alignment increases monotonically with network depth, with the highest correlations emerging in the final layers around the early categorization peak in the EEG (approximately 190–195 ms), consistent with time-resolved RSA and object recognition dynamics reported in prior work (Kriegeskorte et al., 2008; Cichy et al., 2014). This pattern indicates that deeper representations in both streams capture more abstract, temporally precise information that more closely matches the structure of human visual responses during naturalistic video viewing, supporting the idea that the self-selection blocks preserve, and in fact enhance, brain-like representational geometry as information flows through the model (Yamins and DiCarlo, 2016; Schrimpf et al., 2018). The time-course plots in each figure demonstrate that all layers exhibit a clear transient peak near the behavioral decoding maximum, suggesting a consistent latency of category-selective processing across the hierarchy, while the bar plots at the peak make explicit that late layers dominate the overall brain-model correspondence.

The significance matrices further show that differences between early and late layers are statistically reliable (pairwise Wilcoxon signed-rank tests) at the peak latency, particularly between the shallowest and deepest layers in each stream. In the S Where stream, higher layers (5–7) exhibit significantly stronger RSA than the lower ones, reflecting the accumulation of motion and action-relevant features that align with dorsal, spatial processing. We observe a similar progression in the S What stream, but with slightly earlier peak timing and stronger correlations in ventral-leaning regions, consistent with object and appearance-based coding. Together, these statistics show that the sparse selection mechanism not only maintains alignment but also sharpens it in later stages, when the model's internal representations are closest to human EEG.

The scalp topographies at the peak latency, illustrated in Fig. 10 and Fig. 11, visualize how this alignment is distributed across the head for each layer, revealing a qualitative shift from diffuse, low-amplitude correspondence

Comprehensive Layer-Wise Analysis: S What Stream

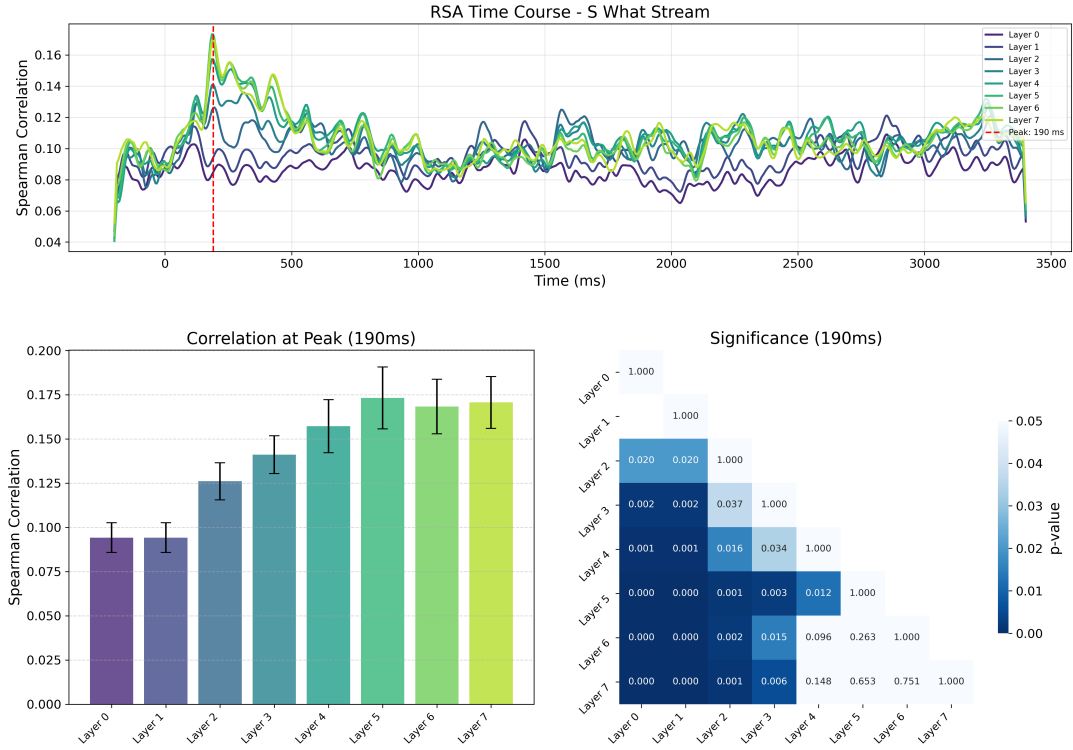


Figure 8: Layer-wise S What stream-EEG RSA analysis. Layer-wise RSA in the S What stream peaks at around 190 ms, with higher layers exhibiting the strongest correlations.

in early layers to more focal, high-amplitude patterns in occipital, temporal, and parietal regions in deeper layers. In the S Where stream, the strongest correlations emerge over parietal and lateral occipital sensors in higher layers, echoing the putative dorsal pathway’s emphasis on spatial relations and motion processing. In the S What stream, late layers show enhanced correlations over ventral occipital and inferior temporal electrodes, in line with ventral visual areas involved in object recognition. For this paper, these figures therefore provide a layer-resolved bridge between the architectural design (separate what and where streams with selection blocks) and the neurobiological motivation, demonstrating that both streams converge on brain-like representations at behaviorally relevant latencies, each with distinct but complementary spatial signatures that mirror ventral and dorsal pathway organization (Goodale and Milner, 1992; Creem and Proffitt, 2001).

B. Appendix: EEG Experiment Details

B.1. Video classes

We selected 21 action sub-classes from the HVU dataset (Diba et al., 2020) to select the videos used in our EEG experiments: ‘riding-or-walking-with-horse’, ‘cutting-the-grass’, ‘swimming’, ‘driving-car’, ‘playing-violin’, ‘boating’, ‘gymnastics’, ‘eating’, ‘motorcycling’, ‘feeding-birds’, ‘opening-present’, ‘playing-flute’, ‘climbing-tree’, ‘drinking’, ‘crawling-baby’, ‘laughing’, ‘playing-drums’, ‘cooking’, ‘petting-cat’, ‘watering-plants’, ‘rollerblading’.

Although each video may contain additional labels (HVU is multi-class and multi-label), we deliberately chose these actions to span a broad semantic space: they include activities involving animate versus inanimate objects and examples of both high-motion and low-motion scenes. We verified that these 21 classes are semantically diverse, using tools like word-embedding similarity measures, so as to cover a wide range of action concepts in our tasks.

Comprehensive Layer-Wise Analysis: S Where Stream

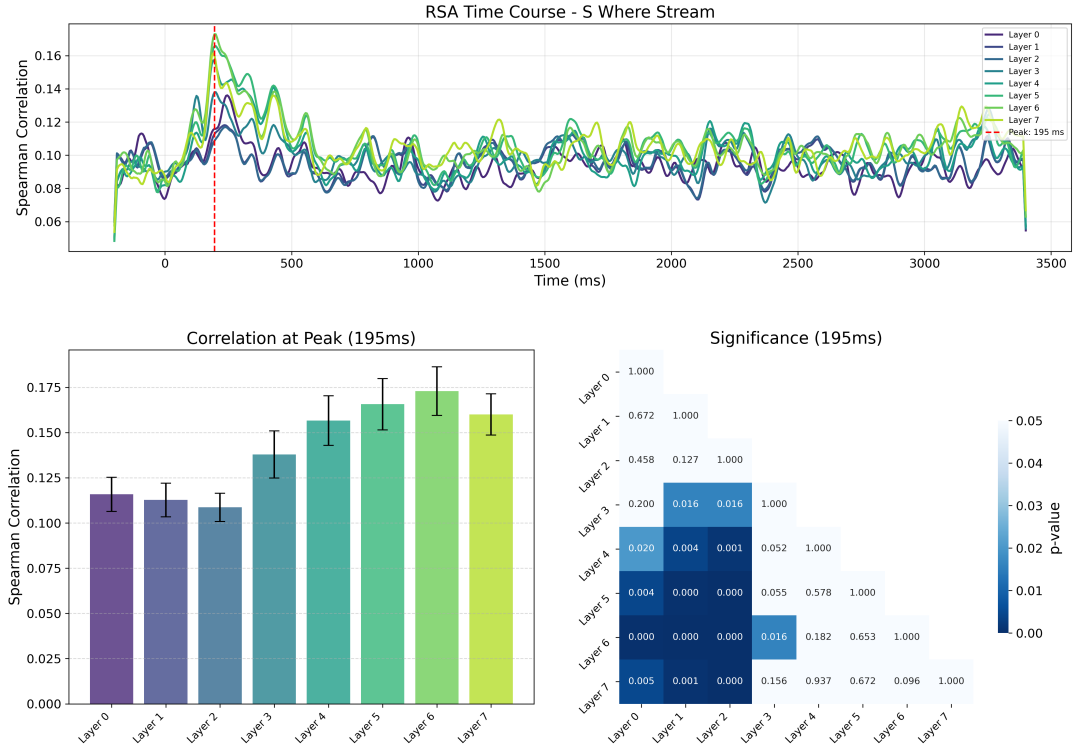


Figure 9: Layer-wise S Where stream-EEG RSA analysis. Layer-wise RSA reveals that deeper S Where layers show progressively stronger alignment with EEG around 195 ms.

B.2. Consent form

To document the ethical safeguards of our EEG experiment, we include a translated version of the participant informed consent form used in the study. This form outlines the purpose of the research, the non-invasive EEG procedures, potential risks and benefits, data confidentiality, compensation, and participants' rights (including voluntary participation and the option to withdraw at any time), and is provided here to facilitate transparent ethical review.

1. Purpose of the Study

I understand that the aim of this study is to compare brain activity in response to static images and dynamic videos using EEG signals. The study seeks to investigate how visual representations in the brain change when transitioning from spatial-only features (static images) to combined spatiotemporal features (videos). By addressing this question, we hope to gain a better understanding of how the brain processes video content, which may inform the development of more accurate and efficient video analysis models. A total of 30 participants will take part in this study, and each individual's contribution is considered valuable in advancing our scientific understanding of brain function.

2. Voluntary Participation

I understand that my participation in this research is entirely voluntary. I am not obligated to participate and will not be denied any medical or diagnostic care if I choose not to take part. My relationship with healthcare providers will not be affected by my decision.

3. Right to Withdraw

I understand that I can withdraw from the study at any time, even after initially agreeing to participate, by informing the researcher. Withdrawing will not affect my access to standard medical services.

4. Participation Procedure

I understand that this study involves EEG recording, a non-invasive and safe neuroimaging method. My participation involves a single session lasting approximately 2 hours during which my brain activity will be recorded while

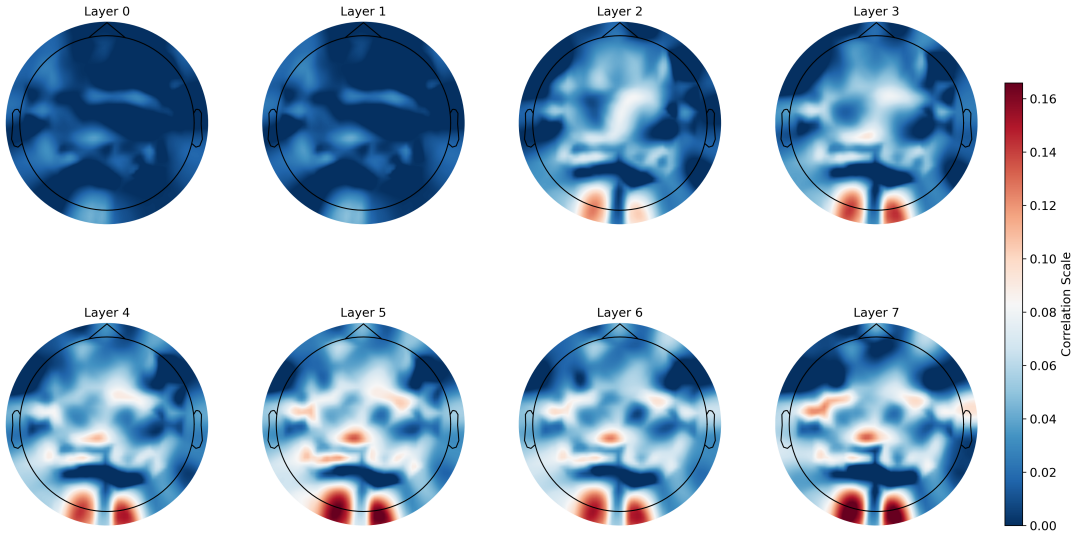


Figure 10: Layer-wise ROI S What stream-EEG RSA analysis at peak timepoint. The later layers show ventral-biased occipital-temporal topographies, consistent with object and appearance-focused ventral processing.

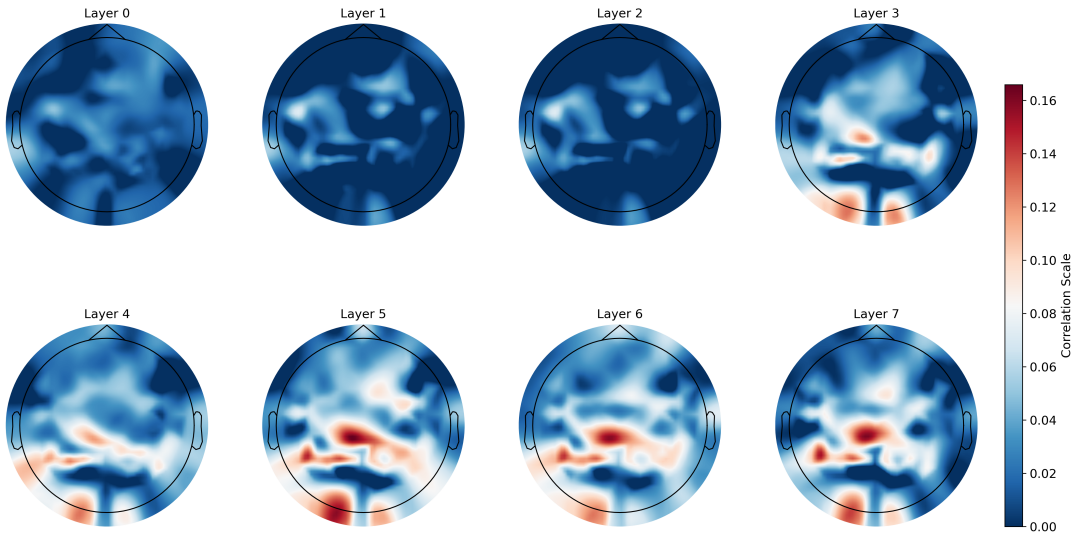


Figure 11: Layer-wise ROI S Where stream-EEG RSA analysis at peak timepoint. Later layers show increasingly focal parietal-occipital topographies, highlighting motion and spatially tuned dorsal-like representations.

viewing stimuli. The session includes several segments, with breaks and refreshments provided in between. An EEG cap with 128 gel-based electrodes will be placed on my head. I will be asked to attentively view a set of video clips and their corresponding static image frames. Reasonable compensation for time and travel expenses will be provided.

5. Potential Benefits

I acknowledge that while there may be no direct personal benefits, a small financial incentive is offered in appreciation of my participation.

6. Risks and Discomforts

I understand that there are no expected risks or side effects associated with EEG recording. This is a completely safe and painless procedure that involves no magnetic fields or harmful exposure. EEG is widely recognized as a safe method for recording brain activity.

7. Confidentiality

I understand that all personal data will be kept strictly confidential. Only group-level results will be published, and no personally identifiable information will be disclosed.

8. Ethics Oversight

I understand that the research ethics committee may access my data to ensure my rights are being respected and that ethical standards are being followed.

9. Costs and Compensation

I understand that I will not incur any costs related to my participation in this research. Compensation for my time and travel expenses will be provided.

10. Contact Information

I was introduced to the [Name] as the point of contact for any questions or concerns I may have regarding the study. His contact information, as well as that of the principal investigator [Name], has been provided to me.

11. In Case of Harm

I understand that if I experience any physical or psychological issues as a result of participating in this study, the research team will be responsible for covering the costs of treatment and compensation.

12. Filing Complaints

I understand that if I have any concerns or complaints about the study or its staff, I can contact the Researches Ethics Committee.

13. Form Copies

I understand that this consent form has been prepared in two copies, one for myself and one for the research team, both of which are to be signed.

B.3. Task instructions

To clarify the behavioral paradigm, the exact task instructions shown to participants on the screen during the EEG recording are reproduced below in translated form. These instructions guided participants through the video and image blocks, explained how to respond to fixation color changes with the space key, and specified when to start, pause, or resume each part of the experiment.

- Introduction

Hello, and thank you very much for taking the time to participate in this experiment. Press the **SPACE** key to continue.

- Video Task Instructions

In this section, a series of videos will be presented. Some of the videos are intact, while in others, the order of frames has been disrupted. Additionally, the color of the + symbol at the center of the screen will occasionally change. Please press the **SPACE** key whenever you notice a color change. Pressing the **SPACE** key will start the video presentation.

- Break

Thank you for staying with us so far. You may now take a short break before continuing with the task. Press the **SPACE** key to resume the video presentation.

- Image Task Instructions

In this section, a series of still images will be presented. Please look at each image carefully. Press the **SPACE** key to start the image presentation.

- End

We sincerely thank you for your participation.

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